

Manual

Archiving Machine Data MDE-ARC 8.2

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Contents

1	Archiving Machine Data	4
2	MDE-specific Configuration.....	6
3	Data Management.....	8
4	Reload Manager.....	12

1 Archiving Machine Data

Purpose

The function package "archiving machine data" not only provides the ability to access archive data, but also to export recorded data and to then upload the data back into the system as needed.

Implementation notes

You use the function package "archiving machine data" if:

- You would like to have access to data already moved to archive tables as part of the function package Machine Data Collection
- Because of legal requirements or customer demands, you need to ensure that the data entered is stored long-term.
- Because of legal requirements or customer demands, you need to import data that was already exported back into the system to be evaluated / analyzed again.

Integration

Machine data archiving accesses recorded data in the respective archive tables of the Machine Data Collection function package. Master data from the Machine Data Collection function package are not archived, but remain in the online tables.

Features

- Direct access to archive tables from the application
 - Direct access to archived machine data from the applications
 - Machine history
 - Machine time profile
 - Efficiency report
 - Performance profile
 - OEE report
 - Status report (workplace/machine-related)
 - Status report
 - Status profile
 - Status class report
 - Status class profile
 - RPA report
 - RPA profile
 - Status analysis

- Transport function
 - Functions transferring data from online tables into archive tables.
- Export function
 - Functions transferring (exporting) data from archive tables to external file systems for the purpose of long-term storage of the recorded machine data.
- Import function
 - Functions importing exported data into archive tables in order to evaluate them using the applications listed above.

2 MDE-specific Configuration

Overview

The data are stored as standard for 35 days in the MDE module before they are moved to long-term storage.

Various MDE evaluations allow data to be accessed which are older than 35 days. For this, the BDE postings are stored in a special medium-term or archive area. Access to these data is essentially automatic as soon as the selection period exceeds the short-term data range. In a few applications there is the option "Consider long-term data" in the selection area via which these data can be accessed.

Data included in the MDE archiving are:

- MDE log records (documents)
- MDE events
- Additional information (e.g. comments)
- LOGGING entries for the machine

Configuration

The retention period of the data in the individual data areas can be configured via the HYDRA data management.



The data management does not define the retention period for data of the "cycle progression" application. The relevant configuration is described in the document dealing with the [cycle progression](#) application.

During the transfer of the data to the archive tables, those data are transferred whose "retention period" (in days/months/years; see values in parentheses in the following table) has been exceeded. If the relevant archiving license for the MDE is not available, the data are deleted after the set retention period.

Product	Object	Object designation	Transfer	Interval in as-delivered condition
MDE	MDEPRO	Log records / documents	Online data → medium-term archive	35 days 1)
MDE	A_MDEPRO	Long-term archiving of log records / documents	Medium-term archive → Long-term archive	3 years 1)

Product	Object	Object designation	Transfer	Interval in as-delivered condition
MDE	EREIGMDE	Events	Online data → medium-term archive	35 days 2)
MDE	A_EREIGMDE	Long-term archiving of events	Medium-term archive → Long-term archive	3 years 2)
MDE	LOG	HYDRA logging data	Online data → medium-term archive	35 days
MDE	A_LOG	Long-term archiving of HYDRA logging data	Medium-term archive → Long-term archive	3 years
MDE	RES_STATUS	Parallel MDE resource status	Online data → medium-term archive	70 days
MDE	A_RES_STATUS	Long-term archiving of parallel MDE resource status	Medium-term archive → Long-term archive	3 years

Please note:

- 1) Please note that the [postings relating to workplaces/machines](#) only allow for data of the online data area to be selected and edited.
- 2) Please note that the [event maintenance](#) only allows for data of the online data area to be selected and edited.

3 Data Management

Overview

Menu	System administration → Archiving → Data management
Transaction code	arccfg
Function authorization	arccfg.*

Purpose

Use this application to view or change the centrally managed [Archiving](#) settings for recorded data.

Integration

Use this application to configure the settings for archiving / data management for all components/functions.

Field descriptions

Product

Enter the HYDRA product group for which you want to define a rule in the *Product* field.

Object

Use the *Object* to define the data you want to retain.

Retention period

Use the fields for the *Retention period* to define how long data should be available until data is archived.

Unit

Specify the retention period in days, months or years.

Last retention date

This field is computed and specifies the last day when data is still available in this object.

Action

You can choose from three options to specify the processing for an object:

D = delete	Object is deleted
M = move (archive)	Object is transferred to the next area
X = export	Object is unloaded (XUNLOAD format) and then deleted

Target object

Currently, the target object automatically results from the detail configuration.

Condition

The characters in this field are added as an additional condition to the database command that controls the action. This condition is linked to the other conditions of the original command using AND and is set in parentheses.

Last run (date/time)

Indicates the point in time when this rule was applied last.

License

Indicates which license is required for archiving. You can enter several licenses (separated by a space). If none of these licenses is licensed in the system, the data is either deleted (action D) or unloaded into files (action M or X), depending on the relevant action.

Path

Optional path to generate the file export (unload). The archiver currently only supports local HYDRA server drives. If you do not enter a path, file exports are filed as follows:

<HYDRADIR>/<SYSTEM>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/

HYDRADIR ... HYDRA directory

SYSTEM ... System number

YYYY-MM-DD ... Archiving date (YYYY ... Year, MM ... Month, DD ... Day)

PRODUCT ... Product group from archiving configuration

Administration table

Name of the administration table where archiving logs are stored.

Administration duration (retention period of administration table)

Use the fields for the *retention period* to specify how long the logs should be available in the administration table before being deleted.

Unit

Specify the retention period in days, months or years.

Archiving step

Specifies if this archiving process uses archiving function I or II. Archiving function I: The function moves data from the online data set to archive tables or deletes the data: setting M (medium-term archive). Archiving function II: The function moves data from the archive table to the file export or deletes the data: setting L (long-term archive).

Configuration

Indicates whether or not the configuration is active. Possible values: Y/N

Archiving type

Identifier for time or object-related archiving. Supported modes: O = Object-related archiving (i.e. data is archived for each object individually). Z = Time-related archiving (i.e. data is archived without any object reference).

Note: Time and object-related archiving differ from each other significantly in the archiving performance (runtime). Object-related archiving of mass data is not recommended.

Priority

Integral value greater than 0. Indicates the processing sequence if several objects are defined for a product group. Processing starts with the lowest value.

Master table

Table including the data to be archived. The extensions entered in the *Condition* field refer to this table.

Date column

Date column in the master table. The system uses this date column to evaluate the retention period for the data to be archived.

Key 1

Unique key column in the master table; this key identifies the data to be archived. You can define up to 5 key columns for object-related archiving. Time-related archiving only supports one primary key in *Key 1*.

Keys 2 – 5

Additional optional key columns for object-related archiving.

Comment

Use the comment line to describe the archiving configuration.



If you copy the archiving configuration, the system currently only copies the data management configuration, but no defined data records from the object details. Currently, you must use the respective dialog to copy the data records of the object details to the new configuration.

Module	Version(s)	Description of archiving
BDE	8.1	See here
CAQ	8.1	See here
LLE	8.1	See here
MDE	8.1	See here
MPL	8.1	See here

PEP	8.1/8.2	See here
PZE/PZW	8.1/8.2	See here
ZKS	8.1/8.2	See here

4 Reload Manager

Summary

HYDRA menu	System administration → Archiving → Reload manager
FEDRA menu	System administration → Archiving → Reload manager
Transaction code	arcrlid
Function authorization	arcrlid arcrlid.export arcrlid.import

Purpose

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evaluate restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

Integration

The Reload Manager is a central function that is used by many components or functions.

Selection Criteria

The application provides the following selection criteria:

Module

Reference to the product group which archived data belongs to.

Object type

Object type of the archived data.



The selection options “module” and “object type” are mandatory fields.

Toolbar

Export

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

Import

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- Cyclic

Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.

Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

When data is imported, the `HYD_REL_MANAGEMENT.DELETE_DATE` field is calculated subject to the configuration and entered in the reload management table.

- User-specific

Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.

When data is imported, the `HYD_REL_MANAGEMENT.DELETE_DATE` field is computed subject to the configuration and entered in the reload management table.

- Manual

Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.